1 Planning (15 marks)

Defining the problem (3 marks)

P *r* is the independent variable or vary *r*. [1]

P *T* (or *t*) is the dependent variable or measure *T* (or *t*). [1]

P Keep the radius of curvature (of the track) or *C* constant (or radius of track constant). Do not allow "use same track". [1]

Methods of data collection (5 marks)

M Diagram showing ball in a (curved) track with supports for track, e.g. retort stands. Minimum of two labels (from ball, track, supports; not stopwatch, bench, micrometer).
Supports making contact with track higher than ball / at least half way up. [1]

M Measure time using stopwatch or light gates and timer or datalogger with motion sensor.
Detail needed for video camera. [1]

M Use many oscillations (at least 10 or at least 10 s of timing) and determine $T = t/n$. [1]

M Measure diameter (radius) of ball with a micrometer / vernier calipers.
Do not allow travelling microscope. [1]

M radius = diameter / 2. [1]

Method of analysis (2 marks)

A Plot a graph of T^2 against *r* (or *r* against T^2)
Do not allow log graphs. [1]

A $C = y\text{-intercept} \times \frac{5g}{28\pi^2} = \frac{\text{y} - \text{intercept}}{\text{gradient}}$ (or for *r* against T^2, $C = y$-intercept) [1]

Safety considerations (1 mark)

S Precaution linked to ball escaping on to floor, e.g. use barrier / safety screen / sand tray to prevent balls rolling on to floor. [1]

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https://xtremepape.rs/